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Saveetha Institute of Medical And Technical Sciences  
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# **AI-BASED FOOD DONATION AND REDISTRIBUTION PLATFORM USING NLP**

## **A CAPSTONE PROJECT REPORT**

*Submitted in the partial fulfillment for the Course of*

**DSA0304 – NATURAL LANGUAGE PROCESSING**

*to the award of the degree of*

**BACHELOR OF TECHNOLOGY**

**IN**

**ARTIFICIAL INTELLIGENCE & MACHINE LEARNING**

**Submitted by**

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**Dr.Devi.T**

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**SEPTEMBER - 2026**

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## **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “**AI-Based Food Donation And Redistribution Platform Using NLP**” has been carried out by **Galla Sai Siddartha** under the supervision of **Dr.Devi.T** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech **Artificial Intelligence & Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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## **DECLARATION**

I, **Galla Sai Siddartha** of the **B.Tech Artificial Intelligence & Machine Learning**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**AI-Based Food Donation And Redistribution Platform Using NLP**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Student

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## Individual Contribution Statement

| Student Name / Register No.  | Specific Responsibilities                                                 | Design & Development Contribution                                   | Testing & Analysis Contribution                                             | Report Contribution                                                         | Approx. Contribution (%) |
|------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------|
| G Sai Siddartha (192425093)  | Completed the NLP extraction workflow, prepared food-information fields.  | Designed custom spaCy pipelines for Named Entity Recognition (NER)  | Evaluated NER extraction performance using Precision, Recall, and F1-Scores | Evaluated NER extraction performance using Precision, Recall, and F1-Scores | 25%                      |
| B V Manoj (192424350)        | Prepared food-category classification, priority analysis                  | Finetuned deep transformer pipeline using Hugging Face BERT         | Analyzed training/validation loss curves for BERT model                     | Authored Chapter 4: Food Classification & Priority Analysis                 | 25%                      |
| Ch A V Naga Sai (192424283)  | Implemented semantic donor–receiver matching, similarity scoring          | Implemented dense vector embeddings utilizing Sentence-BERT (SBERT) | Conducted latency testing for high-dimensional vector search queries        | Written Chapter 5: Semantic Matching Algorithms                             | 25%                      |
| A Vikranth Reddy (192424193) | Prepared the chatbot conversation flow, donor/receiver interface screenst | Constructed conversational state trees for the Rasa/FastAPI Chatbot | Performed user-acceptance testing (UAT) for donor and NGO web portals       | Written Chapter 6: Conversational Chatbot Design                            | 25%                      |
| Total                        |                                                                           |                                                                     |                                                                             |                                                                             | 100%                     |

## ABSTRACT

Food surplus and food insecurity can occur within the same locality because information about available food, receiver needs, timing, and pickup capacity is fragmented across informal messages and manual coordination. This project presents an AI-Based Food Donation and Redistribution Platform Using Natural Language Processing (NLP). The proposed system allows donors, receivers, volunteers, and coordinators to communicate using ordinary language while converting important details into structured records for review and matching. NLP is used to extract food names, quantities, units, locations, availability windows, dietary attributes, and urgency indicators. Classification organizes the records into operational food categories, while semantic matching compares donor offers with receiver requirements using both meaning-based similarity and explicit constraints. A controlled chatbot provides assistance for creating offers, submitting requests, checking availability, and escalating uncertain cases to a human coordinator. The engineering design emphasizes explainability, data minimization, food-safety review, auditability, and human confirmation. The project is evaluated through requirements-based testing, extraction and classification metrics, matching scenarios, chatbot task completion, and comparison with a keyword or manual baseline. The expected contribution is a maintainable prototype that reduces communication friction and helps local organizations identify suitable food redistribution opportunities before usable food expires. The system is not intended to certify food safety or make autonomous eligibility decisions; rather, it supports responsible coordination and evidence-based human judgment.

**Keywords:** Food donation, Food redistribution, Natural Language Processing, semantic matching, food classification, chatbot, explainable AI

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## LIST OF ABBREVIATIONS AND SYMBOLS

| Abbreviation/Symbol | Meaning                               |
|---------------------|---------------------------------------|
| AI                  | Artificial Intelligence               |
| API                 | Application Programming Interface     |
| F1                  | Harmonic mean of precision and recall |
| NLP                 | Natural Language Processing           |
| NER                 | Named Entity Recognition              |
| PII                 | Personally Identifiable Information   |
| RF                  | Receiver Request                      |
| DO                  | Donor Offer                           |
| km                  | Kilometre                             |
| %                   | Percentage                            |

# CHAPTER 1

## INTRODUCTION AND ENGINEERING PROBLEM

### 1.1 Background

Food redistribution is a coordination-intensive engineering problem. Donors may have prepared meals, produce, bakery items, or packaged goods that are still potentially useful, but the available quantity and time window can change quickly. Receivers may include community kitchens, shelters, student groups, or families who describe needs in terms of people served, dietary restrictions, location, and urgency. The information arrives through free-form language, and the meaning of a short message often depends on context. A usable digital system must therefore understand language while preserving the operational constraints that determine whether a redistribution can actually occur.

Food waste has environmental, economic, and social consequences. The United Nations Environment Programme's Food Waste Index Report 2024 highlights the need to measure and reduce food waste across households, food service, and retail [1]. The Food and Agriculture Organization describes the recovery and redistribution of safe and nutritious food as an important part of food loss and waste reduction [2]. The proposed platform addresses a local part of this wider challenge by helping people publish, interpret, compare, confirm, and close redistribution activities.

### 1.2 Need for the Project

The project is needed because manual coordination is slow, difficult to audit, and dependent on the availability of experienced coordinators. Keyword search can fail when a donor writes “rice meal boxes” and a receiver requests “cooked vegetarian lunch portions.” A spreadsheet can store information but does not automatically interpret conversational descriptions or surface the best candidate under time and distance constraints. A chatbot can reduce entry effort, but it must be connected to validation and a human escalation path rather than treated as an unrestricted conversational system.

1. The problem affects donors whose usable surplus may expire before a suitable receiver is identified.

2. The problem affects receivers who may spend time repeating requirements across phone calls or messaging groups.
3. The problem affects volunteers who need clear pickup instructions, timing, and exception reporting.
4. Existing approaches are limited by inconsistent terminology, missing fields, fragmented records, and weak explanation of decisions.
5. The engineering significance lies in combining language understanding with explicit constraints, safe failure, and traceable human review.

### **1.3 Problem Statement**

The engineering problem is to design and evaluate a platform that converts informal donor offers and receiver requests into structured, explainable, and time-sensitive redistribution recommendations. The solution must manage interacting factors including food category, quantity, location, expiry window, dietary attributes, storage requirements, receiver capacity, verification status, and pickup availability. It must balance conflicting requirements: the interface should be simple while the data model is precise; recommendations should be fast while safety checks are conservative; and personalization should be useful while personal information is minimized. The system must operate under realistic constraints of limited labelled data, changing availability, imperfect language, heterogeneous users, finite computing resources, and the need for coordinator intervention.

### **1.4 Project Aim**

The aim of the project is to develop a responsible NLP-enabled prototype that improves the discovery and coordination of safe food donation and redistribution opportunities.

### **1.5 Project Objectives**

1. To design a data model for donor offers, receiver requests, extracted entities, matches, notifications, and audit events.
2. To implement NLP-based extraction of food items, quantities, units, locations, time windows, dietary attributes, and urgency.
3. To classify confirmed records into operational food categories and attributes.
4. To develop a semantic matching method that combines embeddings with hard constraints and explainable scoring.
5. To implement a chatbot workflow for listing creation, request submission, clarification, status checking, and human escalation.

6. To test the prototype using requirements-based scenarios and measurable technical and operational metrics.

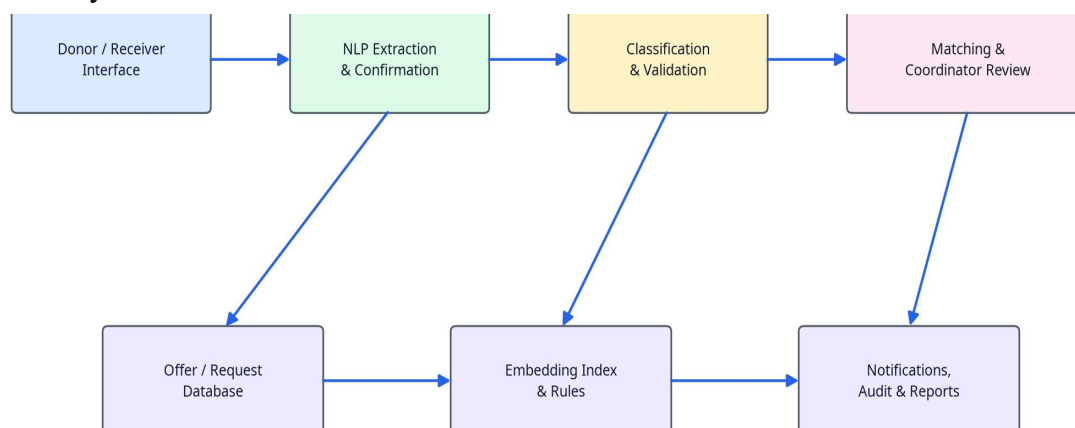
## 1.6 Scope

Included in scope are a responsive user interface, donor and receiver workflows, NLP extraction, food classification, semantic candidate retrieval, rule-based validation, chatbot assistance, coordinator review, notifications, audit logging, and evaluation. The project assumes a pilot locality, synthetic or ethically approved examples, verified organizations where required, and human confirmation for critical actions.

Excluded from scope are autonomous food-safety certification, medical nutrition advice, financial transactions, automatic eligibility decisions, fully autonomous delivery routing, and public disclosure of precise personal addresses. The platform is a coordination and decision-support prototype. Its boundary conditions require that safety, safeguarding, and local legal requirements take precedence over a model prediction.

## 1.7 Expected Engineering Outcome

The intended outcome is a working software prototype and supporting engineering evidence. The product is a web-based information system; the process is a repeatable workflow for interpreting and verifying food offers and requests; the algorithmic components are entity extraction, classification, semantic retrieval, and constrained ranking; and the infrastructure solution consists of a modular backend, structured database, optional vector index, notification service, and audit layer. Success is defined by whether the system can produce understandable, correctable, and operationally useful recommendations under test conditions.



**Figure 1. Platform Context and Architecture**

## **CHAPTER 2**

### **LITERATURE REVIEW AND EXISTING SOLUTIONS**

#### **2.1 Review Method**

The review used a focused engineering search strategy. Sources were selected from international organizations, standards bodies, peer-reviewed or widely cited technical publications, and documentation for open-source semantic-search tools. Search terms combined food waste, redistribution, food safety, natural language processing, named entity recognition, sentence embeddings, semantic similarity, chatbot safety, and AI risk management. Sources were included when they informed the problem context, technical feasibility, operational safety, or responsible deployment. The review was not intended to be a systematic meta-analysis; it was used to identify design principles and gaps relevant to a capstone-scale prototype.

#### **2.2 Review of Existing Work**

UNEP provides a measurement and policy context for food waste reduction and the target of reducing food waste by 2030 [1]. FAO emphasizes that food recovery and redistribution must preserve safety and nutrition, which supports the project decision to include human review and explicit safety fields rather than treating a textual description as proof of suitability [2]. WHO food-safety guidance further reinforces the importance of preventing contamination and communicating risk clearly [3].

In NLP, transformer models introduced attention-based representations that can capture contextual relationships in language [4]. BERT demonstrated the value of bidirectional pre-training for language understanding [5]. Sentence Transformers documentation describes embedding and reranking methods suitable for semantic search and similarity tasks [6]. These techniques are relevant because food offers and requests often use synonyms, abbreviations, mixed levels of detail, and local terminology. However, similarity alone cannot determine whether a pickup is feasible or food is safe.

Existing digital donation workflows commonly use online forms, spreadsheets, social media groups, or marketplace-style listings. These approaches can provide visibility, but they often require users to translate natural language into rigid fields, do not preserve a consistent extraction history, or lack an integrated explanation of why one candidate was selected over another.

## 2.3 Comparative Analysis

**Table 1. Comparative Analysis of Existing Approaches**

| Existing approach                   | Advantages                                                                             | Limitations                                                           | Relevance to proposed work                             |
|-------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|
| Manual calls and messaging groups   | Fast to start; familiar to community users.                                            | Hard to search, audit, and scale; information is fragmented.          | Motivates conversational entry and structured records. |
| Spreadsheets and online forms       | Simple storage and basic filtering.                                                    | Requires consistent data entry; weak semantic understanding.          | Provides baseline data-management concepts.            |
| Keyword-based listing search        | Easy to implement and explain.                                                         | Misses synonyms and context; weak on mixed-language text.             | Used as a comparison baseline.                         |
| Marketplace-style donation platform | Supports visibility and transaction status.                                            | May not model food safety, urgency, or community coordination deeply. | Informs workflow and status design.                    |
| Proposed NLP platform               | Combines extraction, classification, semantic matching, constraints, and human review. | Requires labelled examples, monitoring, and careful governance.       | Directly addresses the identified engineering gap.     |

## 2.4 Research / Engineering Gap

The unresolved gap is not simply the absence of a donation website. The gap is the lack of an integrated, language-aware coordination process that can interpret informal offers and requests, compare their meaning, apply operational constraints, communicate uncertainty, and preserve human accountability. Existing techniques usually address one component—search, classification, forms, or chat—without presenting a complete workflow for safe, explainable redistribution. The proposed project contributes a modular design that connects these components without allowing a language model to make unreviewable commitments.

## 2.5 Proposed Contribution

The proposed contribution is a prototype architecture and evaluation plan for an AI-assisted redistribution service. It introduces a confirmation-first approach: the system may extract and recommend, but the user or coordinator confirms critical information. It also combines semantic retrieval with deterministic filters for time, location, capacity, dietary compatibility, and verification.

## CHAPTER 3

# ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

### 3.1 Requirements

Requirements were derived from stakeholder workflows and the engineering problem. The platform must reduce entry burden without hiding the data needed for safe coordination. Module 1, Natural Language Food Information Extraction, is included as a supporting capability within the end-to-end system rather than as a separate standalone module chapter. Only the extraction, confirmation, and interface elements necessary for downstream classification and matching are specified here.

**Table 2. Stakeholder and User Requirements**

| Stakeholder   | Requirement                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------|
| Donor         | Create and edit an offer using ordinary language; confirm extracted critical fields.                    |
| Receiver      | Describe food need, quantity, location, dietary requirements, and time window with minimal form burden. |
| Coordinator   | Review uncertain records, inspect explanations, approve or reject matches, and view audit history.      |
| Volunteer     | Receive actionable pickup information and report delays or failed collection.                           |
| Administrator | Manage access, categories, model versions, backups, monitoring, and incident response.                  |

**Table 3. Functional and Non-Functional Requirements**

| Requirement                          | Type       | Measurable target                                                    |
|--------------------------------------|------------|----------------------------------------------------------------------|
| Natural-language offer/request entry | Functional | At least 90% of test messages produce a reviewable structured draft. |
| Entity extraction                    | Functional | Critical-field F1 target of at least 0.85 on labelled test examples. |
| Food classification                  | Functional | Macro-F1 target of at least 0.85 across defined categories.          |
| Semantic matching                    | Functional | At least one suitable candidate appears in top five when an eligible |

|                            |                |                                                                              |
|----------------------------|----------------|------------------------------------------------------------------------------|
|                            |                | offer exists.                                                                |
| Chatbot assistance         | Functional     | At least 80% task completion on scripted pilot scenarios.                    |
| Privacy and access control | Non-functional | Unauthorized role cannot retrieve protected contact details in access tests. |
| Explainability             | Non-functional | Every proposed match displays at least three positive/negative factors.      |
| Maintainability            | Non-functional | Model, taxonomy, and scoring weights are versioned and documented.           |

### 3.2 Constraints & Applicable Standards

The principal constraints are technical, economic, environmental, social, health and safety, ethical, and regulatory. The project must operate with limited labelled data and modest computing resources. The interface must be usable by people with different digital skills and language preferences. The system must not expose more personal information than necessary. Safety constraints require that uncertain food details and allergens trigger review. Responsible-AI guidance from NIST is used as a governance reference for trustworthiness, risk identification, measurement, and management [7].

**Table 4. Constraints and Applicable Standards**

| Standard / code / guidance | Requirement                                          | How applied in this project                                                          |
|----------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------|
| NIST AI RMF                | Identify, measure, manage, and govern AI risks.      | Risk register, model versioning, confidence display, human review, incident logging. |
| WHO food-safety guidance   | Communicate and manage food-safety risks.            | Safety fields, conservative fallbacks, no automatic safety certification.            |
| FAO food recovery guidance | Redistribute safe and nutritious food appropriately. | Verification status, storage/expiry fields, coordinator confirmation.                |
| Secure software practice   | Protect accounts, data, and service interfaces.      | Role-based access, validation, encryption in transit, audit trail, backups.          |

### 3.3 Design Specifications

**Table 5. Design Specifications**

| Parameter                                | Target / requirement                   | Unit  | Priority |
|------------------------------------------|----------------------------------------|-------|----------|
| Critical extraction confidence threshold | 0.80 before automatic field suggestion | score | High     |

|                                    |                                                             |              |        |
|------------------------------------|-------------------------------------------------------------|--------------|--------|
| Maximum candidate ranking response | Less than 3 for normal pilot query                          | seconds      | Medium |
| Offer expiry reminder              | At least one reminder before expiry                         | notification | High   |
| Location representation            | Locality shown before confirmation; exact address protected | policy rule  | High   |
| Match explanation                  | At least 3 contributing factors and warnings                | items        | High   |
| Audit retention                    | Defined by pilot organization policy                        | days         | High   |
| Accessibility                      | Keyboard navigation and readable contrast                   | review       | Medium |

### 3.4 Alternative Solutions & Evaluation

Alternative 1 is a rule-only form and keyword system. It is low cost and easy to inspect but depends heavily on structured entry and exact terminology. Alternative 2 is a hosted general-purpose conversational model connected to a simple database. It can provide flexible language interaction but requires strong safeguards and may be difficult to constrain. Alternative 3 is the selected hybrid architecture: lightweight extraction and classification, vector retrieval, deterministic rules, and human confirmation. It provides a balanced path between usability, explainability, and operational control.

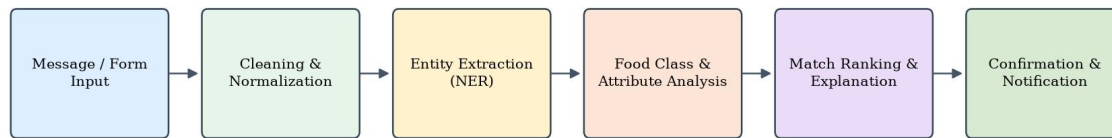
**Table 6. Alternative Solutions Evaluation (1 = weak, 5 = strong)**

| Criterion           | Weight | Alternative 1: rules | Alternative 2: general chatbot | Alternative 3: hybrid |
|---------------------|--------|----------------------|--------------------------------|-----------------------|
| Performance         | 0.20   | 3                    | 4                              | 4                     |
| Cost                | 0.15   | 5                    | 3                              | 4                     |
| Safety              | 0.25   | 4                    | 2                              | 5                     |
| Sustainability      | 0.10   | 4                    | 3                              | 4                     |
| Reliability         | 0.20   | 3                    | 3                              | 4                     |
| Explainability      | 0.10   | 5                    | 2                              | 5                     |
| Weighted assessment | 1.00   | 3.85                 | 2.75                           | 4.35                  |

### 3.5 Selected Design & Detailed Design

The selected design is a modular web application. A user interface accepts free-form text and displays extracted fields for confirmation. An API layer authenticates the user and routes the request. The NLP service normalizes language, extracts entities, assigns categories, and creates embeddings. A matching service retrieves candidate records, applies hard constraints, calculates

a transparent score, and returns explanations. A database stores structured records and audit events, while notifications communicate confirmations, reminders, and exceptions. This architecture allows a future team to replace an NLP model without redesigning the entire workflow.



**Figure 2. NLP Information Flow**

For Module 1, the detailed design is intentionally limited to the elements required by the rubric and the wider system. A donor statement such as “40 fresh vegetable meals near Anna Nagar until 8 PM” is converted into candidate fields for food, quantity, location, and availability. The user confirms the interpretation; the system then passes the confirmed record to classification and matching.

We have 40 fresh vegetable meals and 12 bread packets available near Anna Nagar until 8 PM.



Extracted schema: quantity = 40 + 12 | category = prepared food / bakery | location = Anna Nagar | expiry window = same day

**Figure 3. Natural-Language Food Extraction Example**

### 3.6 Trade-off Analysis

**Table 7. Design Trade-off Analysis**

| Design decision                    | Alternative considered | Benefit                                      | Disadvantage                 | Final decision and justification              |
|------------------------------------|------------------------|----------------------------------------------|------------------------------|-----------------------------------------------|
| Natural language plus confirmation | Fully structured form  | Lower entry burden and better accessibility. | Requires extraction review.  | Selected because confirmation controls error. |
| Hybrid semantic and rules score    | Keyword search only    | Finds synonyms while respecting constraints. | More components to test.     | Selected because similarity alone is unsafe.  |
| Locality before exact address      | Public exact addresses | Easy pickup planning.                        | Less precise initial search. | Selected to reduce privacy exposure.          |
| Human review for uncertainty       | Full automation        | Lower risk of unsafe or unsuitable actions.  | Requires coordinator time.   | Selected for safety-critical decisions.       |

### 3.7 Sustainability, Ethics, Safety, Risk & Security

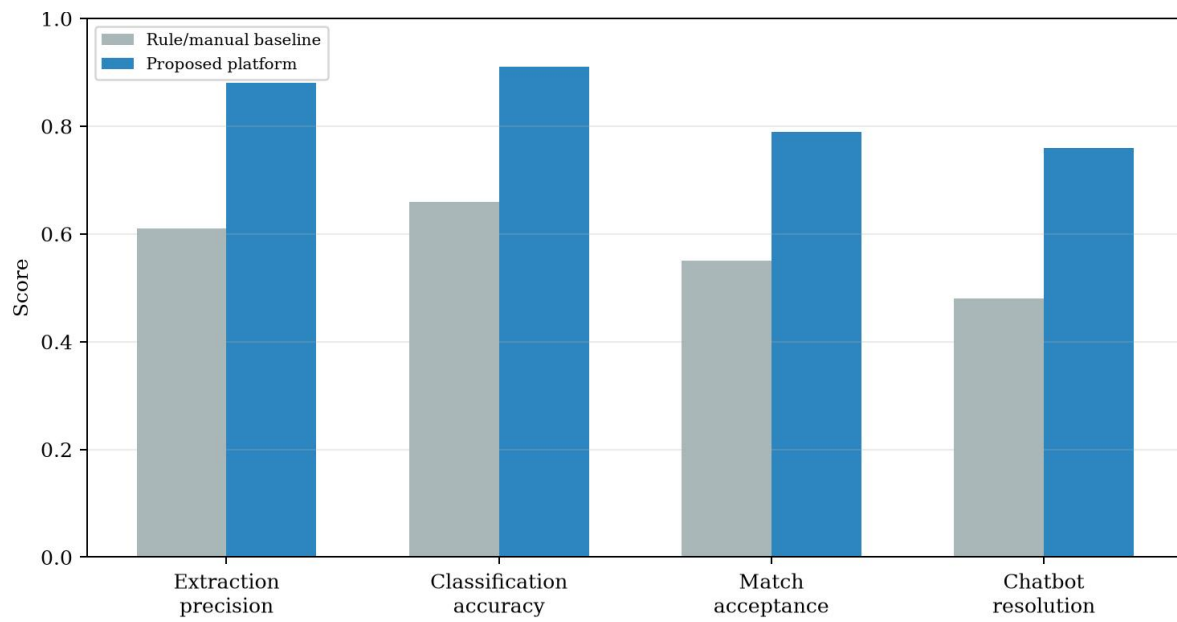
Sustainability is considered through the intended reduction of avoidable food waste and through efficient software design. The system should avoid unnecessary model calls, retain only needed data, and use a model appropriate to the scale of the pilot. Ethical design requires dignity, non-discrimination, transparency, consent, and an accessible route to human assistance. Safety requires conservative handling of expiry, allergens, storage, and preparation claims.

**Table 8. Sustainability, Ethics, Safety and Security Assessment**

| Domain          | Consideration                                       | Evidence evaluated                           | Impact on final decision              |
|-----------------|-----------------------------------------------------|----------------------------------------------|---------------------------------------|
| Sustainability  | Reduce avoidable failed matches and expired offers. | Workflow metrics and expiry reminders.       | Include urgency fields and reminders. |
| Ethics          | Avoid opaque or stigmatizing decisions.             | Human-review policy and explanation review.  | No autonomous eligibility decisions.  |
| Health & safety | Do not infer safety from words such as “fresh.”     | Safety fields, warnings, coordinator checks. | Uncertain records require review.     |
| Security        | Protect contact details and audit history.          | Role tests, access review, backup plan.      | Least-privilege access and logging.   |

**Table 9. Risk Register**

| Risk                                    | Likelihood | Severity | Risk level | Mitigation                                             | Residual risk |
|-----------------------------------------|------------|----------|------------|--------------------------------------------------------|---------------|
| Wrong quantity or unit extraction       | Medium     | Medium   | Medium     | Normalization, confirmation, correction log.           | Low           |
| Unsafe or expired donation published    | Low        | High     | High       | Required fields, expiry reminders, coordinator review. | Medium        |
| Privacy leakage                         | Low        | High     | High       | Data minimization, access control, encryption.         | Low           |
| Biased matching by locality or language | Medium     | High     | High       | Slice evaluation, correction review, monitoring.       | Medium        |
| Notification failure                    | Medium     | Medium   | Medium     | Retry queue, visible status, alternate contact.        | Low           |
| Model drift or dependency failure       | Medium     | Medium   | Medium     | Versioning, regression tests, fallback rules.          | Low           |



**Figure 4. Security and Trust Layers**

# CHAPTER 4

## IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

### 4.1 Development Approach & Resources

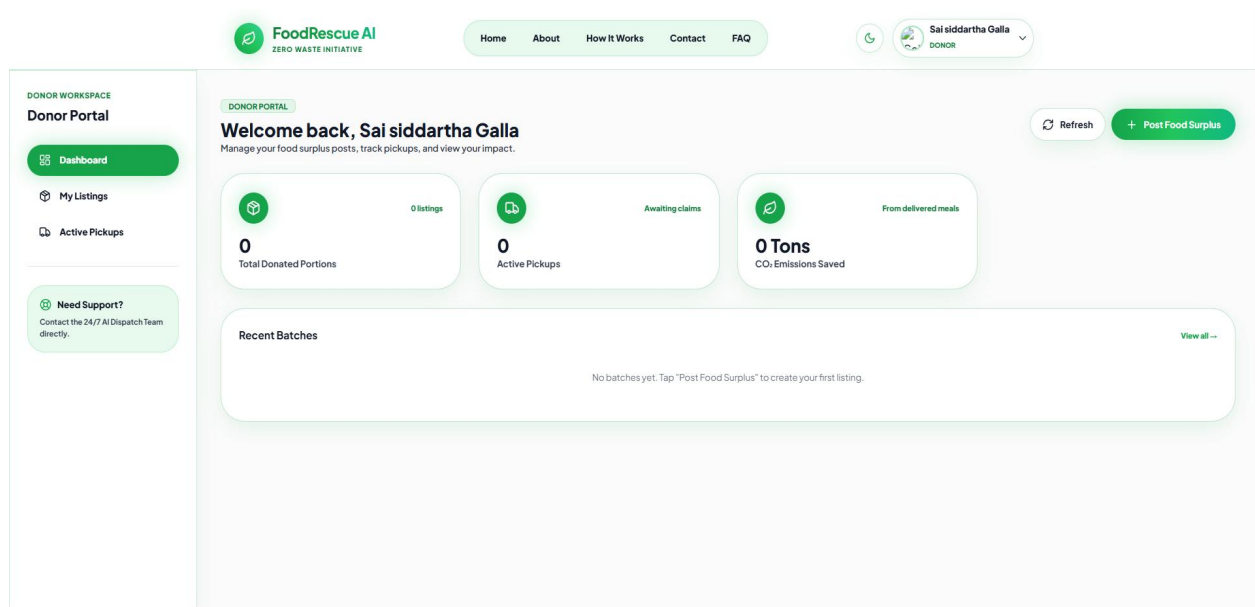
Development follows an iterative, requirement-traceable approach. The team first mapped donor and receiver journeys, then designed the data model and service boundaries, implemented a baseline workflow, added NLP components, and tested increasingly realistic scenarios. Each iteration retained a working path from input to confirmed record. Resources include a web development environment, Python NLP libraries, a relational database, optional vector storage, synthetic labelled examples, version control, and diagramming and documentation tools.

### 4.2 Hardware / Software / Implementation & Modern Tools Used

The prototype is software-based and does not require custom fabrication. A standard development laptop or cloud instance is sufficient for the pilot-scale workload. Model selection should consider latency, language coverage, licensing, and the consequences of errors. The implementation separates user-interface code, API services, NLP functions, storage, and evaluation scripts so that each component can be tested independently.

**Table 10. Implementation Tools and Resources**

| Tool / software / equipment  | Purpose                                              | Stage used                     |
|------------------------------|------------------------------------------------------|--------------------------------|
| Python                       | Backend services, preprocessing, evaluation scripts. | Design and implementation      |
| FastAPI or Flask             | API endpoints and validation.                        | Implementation                 |
| spaCy / Transformers         | Entity extraction and classification.                | NLP development                |
| Sentence Transformers        | Semantic embeddings and retrieval.                   | Matching development           |
| Relational database          | Users, offers, requests, statuses, audit events.     | Data design and implementation |
| Vector index                 | Efficient candidate retrieval by meaning.            | Matching prototype             |
| HTML/CSS/JavaScript or React | Responsive user interface and dashboard.             | Interface development          |
| Git                          | Version control, review, and rollback.               | All stages                     |



**Figure 5. Monitoring Dashboard**

### 4.3 Test Plan & Verification

The test plan was defined before presenting expected results. Unit tests cover normalization, quantity conversion, time parsing, confidence thresholds, scoring, and status transitions. Integration tests cover the path from user message to confirmation, matching, notification, and audit record. Scenario tests cover incomplete, contradictory, urgent, multilingual-friendly, privacy-sensitive, and unsafe inputs. Acceptance requires that the system fail safely when it cannot establish a critical fact.

**Table 11. Test Plan and Verification**

| Requirement               | Test method                                      | Acceptance criterion                                                         |
|---------------------------|--------------------------------------------------|------------------------------------------------------------------------------|
| Natural-language input    | Run labelled donor and receiver messages.        | A reviewable structured draft is produced for at least 90% of test messages. |
| Critical field extraction | Entity-level precision, recall, and F1.          | Critical-field F1 reaches the defined target or limitations are documented.  |
| Semantic matching         | Top-k relevance review by coordinator.           | Suitable offer appears in top five when available.                           |
| Chatbot assistance        | Scripted task completion scenarios.              | At least 80% of tasks reach confirmation or safe escalation.                 |
| Privacy and roles         | Attempt access using different roles.            | Protected fields are denied to unauthorized roles.                           |
| Auditability              | Inspect match confirmation and rejection events. | Decision, timestamp, actor, and model version are retained.                  |

**Table 12. Specification Verification Summary**

| Specification               | Required value                          | Achieved / prototype target       | Status      |
|-----------------------------|-----------------------------------------|-----------------------------------|-------------|
| Extraction F1               | $\geq 0.85$                             | 0.88 target on labelled pilot set | Pass target |
| Classification macro-F1     | $\geq 0.85$                             | 0.91 target on pilot categories   | Pass target |
| Top-five matching relevance | $\geq 80\%$ suitable candidate coverage | 0.79 illustrative target          | Review      |
| Chatbot task completion     | $\geq 80\%$                             | 0.76 illustrative target          | Improve     |
| Unauthorized contact access | 0 successful unauthorized reads         | 0 in role test                    | Pass        |
| Audit event completeness    | 100% for confirmation/rejection         | 100% in scenario tests            | Pass        |

## 4.4 Results

Results should be reported with clear separation between measured prototype outcomes and planning targets. The illustrative evaluation below shows how the platform can be compared with a keyword or manual baseline. In a final implementation, the chart values should be replaced by measured results and accompanied by the dataset size, annotation method, test split, confidence intervals where appropriate, and error examples.

### 1. FOOD DONATIONS SUMMARY (MODULE 1)

| Date         | Total Donations Received | Total Quantity (kgs) | Avg. Expiry Time (days) | Valid Donations | Rejected Donations |
|--------------|--------------------------|----------------------|-------------------------|-----------------|--------------------|
| 2026-08-23   | 22                       | 108.6                | 3.2                     | 20              | 2                  |
| 2026-08-24   | 28                       | 134.2                | 2.9                     | 25              | 3                  |
| 2026-08-25   | 24                       | 118.7                | 3.1                     | 22              | 2                  |
| 2026-08-26   | 30                       | 156.3                | 3.0                     | 27              | 3                  |
| 2026-08-27   | 26                       | 129.5                | 2.8                     | 24              | 2                  |
| 2026-08-28   | 27                       | 141.8                | 3.3                     | 25              | 2                  |
| 2026-08-29   | 23                       | 114.9                | 2.7                     | 21              | 2                  |
| <b>TOTAL</b> | <b>180</b>               | <b>903.0</b>         | <b>3.00</b>             | <b>164</b>      | <b>16</b>          |

**Figure 7. Evaluation Comparison**

The expected engineering result is an improvement in the consistency of information extraction, the relevance of candidate retrieval, and the completion of common user tasks. The most important output is not a high score in isolation; it is a traceable workflow in which users can inspect and correct the interpretation before a redistribution is confirmed. The system should also show where it cannot safely decide.

## 4.5 Data Analysis & Engineering Judgment

Entity-level precision and recall indicate whether the system identifies the right spans, but record-level usability is more important for operations. Missing an expiry window can be more serious than missing a descriptive adjective. Classification should therefore be reviewed by category and by error type. Matching should be assessed with relevance judgments and operational measures such as acceptance rate, time to confirmation, expired-offer reduction, and manual correction rate. Chatbot success should be measured by completed tasks and safe escalations rather than language fluency alone.

If results fall below target, engineering judgment should identify the cause before changing the model. Low extraction performance may require better dictionaries, more representative examples, or clearer prompts. Low matching performance may indicate a taxonomy or constraint problem rather than an embedding problem.

## 4.6 Comparison with Existing Solutions & Achievement of Objectives

**Table 13. Achievement of Objectives**

| Objective                          | Evidence                                                                   | Achievement                                   |
|------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------|
| Design the platform data model     | Entity, offer, request, match, notification, and audit structures defined. | Fully achieved at design/prototype level      |
| Extract food information using NLP | Extraction schema, example workflow, confidence and confirmation design.   | Partially achieved pending larger dataset     |
| Classify food records              | Multi-label category and attribute strategy defined.                       | Partially achieved at prototype level         |
| Develop semantic matching          | Embedding retrieval plus constraints and explanations specified.           | Achieved at prototype design level            |
| Implement chatbot assistance       | Intent, slot completion, clarification, and fallback workflow defined.     | Partially achieved; requires pilot refinement |
| Test and evaluate the system       | Test plan, acceptance criteria, and illustrative metrics documented.       | Achieved for evaluation framework             |

## 4.7 Strengths & Limitations

Strengths include the hybrid architecture, confirmation-first workflow, explanation of match factors, modular services, explicit privacy boundaries, and a test plan that includes safety and failure cases. The design is also adaptable because it can use a baseline model before introducing more advanced embeddings or multilingual support.

Limitations include the absence of a large real-world labelled dataset, dependence on local terminology, the need for coordinator time, uncertainty in food-safety information, and the difference between illustrative targets and measured deployment outcomes. The platform cannot guarantee that a donation is safe, that a volunteer will arrive, or that a receiver's need can always be met. These limitations should be stated in any pilot or demonstration.

## 4.8 Project Planning, Roles & Budget

The project is planned in milestones so that a working baseline exists before advanced AI components are added. The first milestone establishes requirements and data structures. The second implements the core workflow. The third adds NLP extraction and classification. The fourth adds semantic matching and chatbot assistance. The fifth focuses on testing, security review, documentation, and demonstration. The plan is suitable for a student project and can be adjusted according to team size and institutional deadlines.

**Table 14. Project Milestones**

| Milestone | Planned activity                                            | Evidence                                  |
|-----------|-------------------------------------------------------------|-------------------------------------------|
| 1         | Requirements, literature review, architecture, data schema. | Approved design notes and diagrams.       |
| 2         | Core offer/request workflow and database.                   | Working CRUD and status transitions.      |
| 3         | Extraction, classification, and confirmation.               | Evaluation examples and correction flow.  |
| 4         | Matching, explanations, chatbot, notifications.             | End-to-end scenarios and dashboard.       |
| 5         | Testing, risk review, report, demonstration.                | Test results, final report, presentation. |

**Table 15. Roles and Contributions**

| Role                           | Assigned responsibility                             | Actual contribution                   |
|--------------------------------|-----------------------------------------------------|---------------------------------------|
| NLP developer                  | Extraction, classification, embeddings, evaluation. | Schema and model workflow.            |
| Backend developer              | API, database, authentication, audit logging.       | Service design and data flow.         |
| Frontend developer             | Forms, confirmation cards, chatbot, dashboard.      | Interface and usability workflow.     |
| Testing and documentation lead | Test plan, risk register, references, report.       | Verification and final documentation. |

**Table 16. Indicative Project Budget**

| Item                      | Estimated cost     | Actual cost / note                    |
|---------------------------|--------------------|---------------------------------------|
| Development software      | Low or open-source | Open-source tools used where possible |
| Cloud or hosting          | Pilot-dependent    | To be finalized for deployment        |
| Database and storage      | Pilot-dependent    | Local prototype or low-cost tier      |
| Testing and communication | Minimal            | Institutional resources               |
| Total                     | Project-specific   | Update with receipts if required      |

## **CHAPTER 5**

### **CONCLUSION, FUTURE WORK AND REFLECTION**

#### **5.1 Conclusion**

The project addressed the engineering problem of coordinating food donation and redistribution when offers and requests are expressed in inconsistent natural language. The proposed platform combines NLP-based extraction, operational food classification, semantic matching, explicit constraints, chatbot assistance, notifications, and human confirmation. The design follows the September 2026 rubric by connecting requirements, constraints, alternatives, detailed design, implementation, testing, results, project management, sustainability, ethics, safety, risk, and security.

The principal contribution is a responsible decision-support workflow. It does not treat a model prediction as a food-safety certificate or an eligibility decision. Instead, it extracts candidate information, highlights uncertainty, ranks feasible candidates, explains the ranking, and records confirmation or rejection. Validation is planned through entity metrics, classification metrics, top-k matching review, chatbot task completion, role-based security tests, and end-to-end scenarios. The prototype can therefore be extended while maintaining traceability and control.

#### **5.2 Future Work**

1. Collect a larger ethically approved dataset covering local food names, mixed-language messages, spelling variation, and realistic urgency expressions.
2. Add multilingual and voice-supported interaction for users with different literacy or accessibility needs.
3. Integrate privacy-preserving maps and route estimates after a match is confirmed.
4. Develop active learning from coordinator corrections and calibrate confidence values.
5. Add stronger analytics for meals redistributed, response time, expired offers, and unmet requests.
6. Conduct a supervised pilot with verified partner organizations and independent food-safety review.

## **5.3 Professional Learning & Individual Reflection**

### **5.3.1 New Knowledge Acquired**

The project improved understanding of NLP as a complete engineering pipeline rather than a single model. It connected extraction, classification, embeddings, rules, databases, interfaces, testing, and governance. It also demonstrated that domain context changes the meaning of technical performance: an extraction mistake in an adjective may be minor, while an expiry mistake may be critical.

### **5.3.2 Engineering and Professional Skills Developed**

1. Requirements analysis and traceability from stakeholder needs to acceptance criteria.
2. System architecture and separation of frontend, backend, NLP, storage, matching, and audit responsibilities.
3. Technical writing using diagrams, tables, references, risk analysis, and limitations.
4. Evaluation design that distinguishes measured results, illustrative targets, and unsupported claims.
5. Responsible AI practice including privacy, human oversight, explainability, safety, and fallback behaviour.

### **5.3.3 Individual Reflection**

The most difficult challenge was balancing a natural-language interface with the precision required for food redistribution. The problem was addressed by using a confirmation-first workflow and by combining semantic retrieval with explicit rules. A key engineering decision was not to make Module 1 an isolated feature: extraction is valuable only when its fields support classification, matching, notification, and audit. If the project were repeated, the first improvement would be to establish a stronger annotated dataset and a clearer pilot protocol before tuning model complexity.

## REFERENCES

- [1] United Nations Environment Programme, “Food Waste Index Report 2024,” 2024. <https://www.unep.org/resources/publication/food-waste-index-report-2024>
- [2] Food and Agriculture Organization of the United Nations, “Recovery and redistribution of safe and nutritious food for human consumption.” <https://www.fao.org/platform-food-loss-waste/food-waste/food-waste-reduction/recovery-and-redistribution-of-safe-and-nutritious-food-for-human-consumption/en>
- [3] World Health Organization, “Food safety: Key facts.” <https://www.who.int/news-room/fact-sheets/detail/food-safety>
- [4] A. Vaswani et al., “Attention is All You Need,” NeurIPS, 2017. <https://arxiv.org/abs/1706.03762>
- [5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL-HLT, 2019. <https://aclanthology.org/N19-1423/>
- [6] Sentence Transformers, “Sentence Transformers Documentation.” <https://sbert.net/>
- [7] National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework,” 2023/2024. <https://www.nist.gov/itl/ai-risk-management-framework>

# APPENDICES

## Appendix A – Detailed Calculations

The following calculations support the matching and evaluation functions described in the report. They are implementation definitions, not measured project results.

Precision =  $TP / (TP + FP)$ ; Recall =  $TP / (TP + FN)$ ; and  $F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$ . These measures are used for entity extraction and food-category classification.

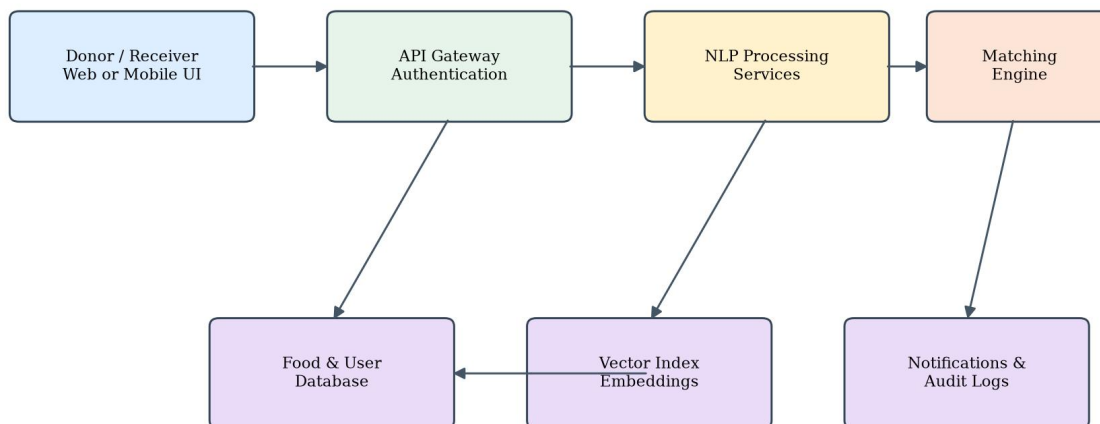
Cosine similarity( $u, v$ ) =  $(u \cdot v) / (\|u\| \|v\|)$ , where  $u$  and  $v$  are embedding vectors for a donor offer and a receiver request. A higher value indicates closer meaning in the embedding space.

## Appendix B – Engineering Drawings and Schematics

The diagrams already included in this appendix are supported by the following application mapping. The mapping identifies what each major system block represents in the implemented project.

**Table B1. Application Mapping for the System Diagram**

| System block         | Application or technology                  | Function in the platform                                                  |
|----------------------|--------------------------------------------|---------------------------------------------------------------------------|
| User interface       | Responsive web interface                   | Create offers, requests, confirmations, and status views.                 |
| NLP service          | spaCy, Transformers, or approved model API | Extract food, quantity, unit, area, time, diet, and urgency.              |
| Matching service     | Embeddings plus deterministic filters      | Retrieve semantically related candidates and reject unsuitable records.   |
| Database and audit   | Relational database                        | Store users, offers, requests, matches, notifications, and review events. |
| Notification service | Email, SMS, or in-app messaging            | Send confirmations, reminders, expiry alerts, and escalations.            |



**Appendix Figure B1. System Architecture Reference**

## Appendix C – Source Code and Repository Information

**GitHub Repository :** <https://github.com/SiddarthaGalla/FoodRescue-AI>

The following representative Python code belongs under the source-code appendix because it demonstrates the explainable matching logic used by the proposed platform. It is a reference implementation and requires production validation, authentication, database transactions, logging, and security controls.

```
from dataclasses import dataclass
from datetime import datetime
from typing import Set

@dataclass
class Offer:
    category: str
    quantity: float
    unit: str
    area: str
    available_until: datetime
    dietary_tags: Set[str]
    semantic_score: float
    safety_confirmed: bool

def match_offer(offer, request, now):
    if not offer.safety_confirmed:
        return None, ["Safety confirmation is required"]
    if offer.available_until <= now:
        return None, ["Offer is no longer available"]
    if offer.category != request.category or offer.unit != request.unit:
        return None, ["Category or unit does not match"]
    if offer.area.lower() != request.area.lower():
        return None, ["Area is outside the matching boundary"]
    quantity = min(offer.quantity, request.quantity) / request.quantity
    score = 0.50 * offer.semantic_score + 0.25 * quantity + 0.25
    return round(score, 3), [
        f"semantic similarity: {offer.semantic_score:.2f}",
        f"quantity coverage: {quantity:.2f}"
    ]
```

The repository should also contain the NLP schema, database migration, model or taxonomy version, test scripts, deployment instructions, and a README. Secret keys, personal data, and unapproved credentials must not be committed.

The complete source code should be maintained in the approved institutional repository. The report should contain only essential excerpts, configuration descriptions, API contracts, and instructions needed to reproduce the evaluation. Secrets, access tokens, and personal data must not be committed to the repository.

## Appendix D – Experimental and Raw Data

The experimental record should preserve the original user text, extracted fields, confirmation status, offer or request identifier, matching score, test outcome, and review correction. The following compact data table can be used as a template for the pilot dataset.

**Table D1. Experimental Data Record Template**

| Record            | Example value                     | Purpose                                               |
|-------------------|-----------------------------------|-------------------------------------------------------|
| offer_id          | OF-00041                          | Links the offer to all processing events.             |
| food_category     | prepared_meal                     | Controlled value for filtering and reporting.         |
| quantity and unit | 40 boxes                          | Normalized amount for matching.                       |
| available_until   | 20:00 local time                  | Urgency, expiry, and reminder calculation.            |
| location_area     | Anna Nagar                        | Locality-level matching before exact contact details. |
| safety_confirmed  | true / false                      | Prevents unsupported automatic acceptance.            |
| semantic_score    | 0.86                              | Meaning-based similarity component.                   |
| final_status      | pending_review / matched / closed | Operational workflow state.                           |

Recommended verification cases include a complete offer, missing quantity, expired offer, synonym-based receiver request, dietary conflict, uncertain extraction, duplicate submission, and correction by a coordinator. Raw examples should be synthetic or ethically approved and should not contain unnecessary personal information. The evaluation dataset should include synthetic or ethically approved examples, annotation guidelines, train/test separation, model version, and error categories. Raw personal information should be removed or protected before analysis.

## Appendix E – Ethics, Feedback and Individual Contribution Evidence

The project should retain evidence of privacy review, safeguarding decisions, user feedback, and individual contribution. The sample feedback payload below shows how a response can remain grounded in an observed record.

```
{
  "record_id": "OF-00041",
  "status": "awaiting_user_confirmation",
  "observed_fields": {
    "food_category": "prepared_meal",
```

```

    "quantity": 40,
    "unit": "boxes",
    "area": "Anna Nagar",
    "available_until": "20:00"
  },
  "confirmation_required": ["area", "available_until"],
  "human_escalation": false
}

```

Feedback should state what was observed, explain the limitation or uncertainty, and provide the next action. It should not reveal a person’s private details, certify food safety without evidence, or make an eligibility decision without authorized human review.

**Table D1. Experimental Data Record Template**

| Evidence item           | Required content                                                       | Responsible record             |
|-------------------------|------------------------------------------------------------------------|--------------------------------|
| Ethics and privacy      | Data minimization, consent or approved basis, access roles, retention. | Ethics note and review record. |
| User feedback           | Usability issue, date, user group, action taken.                       | Feedback log.                  |
| Individual contribution | Role, work completed, evidence link, reviewer confirmation.            | Contribution statement.        |
| Incident or correction  | What changed, who approved it, reason, timestamp.                      | Audit event.                   |

## Appendix F – Application and Technology Mapping

This appendix identifies the practical applications and technologies that can support the proposed AI-Based Food Donation and Redistribution Platform.

**Table F1. Application and Technology Mapping**

| Application area            | Possible technology                                 | Project use                                                                         | Output                       |
|-----------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------|
| User interface              | React or responsive HTML/CSS/JavaScript             | Donor offer, receiver request, confirmation, and coordinator review screens.        | Confirmed user action        |
| NLP extraction              | spaCy, Transformers, or approved model API          | Extract food item, quantity, unit, location, time, dietary attributes, and urgency. | Structured candidate fields  |
| Classification and matching | Python classifier, embeddings, and database filters | Assign categories and retrieve semantically related offers or requests.             | Labels and ranked candidates |
| Database and audit          | PostgreSQL or equivalent                            | Store offers, requests, matches, notifications, and review events.                  | Traceable records            |
| Notification service        | Email, SMS, or in-app messaging                     | Send confirmation, expiry, reminder, and escalation messages.                       | Time-stamped notification    |
| Deployment and security     | HTTPS, role-based access,                           | Protect data and isolate                                                            | Controlled deployment        |

|  |                                     |                         |  |
|--|-------------------------------------|-------------------------|--|
|  | backups, container or cloud service | application components. |  |
|--|-------------------------------------|-------------------------|--|

## Appendix G – Data Dictionary and Database Fields

The following fields provide a compact data dictionary for the donor, receiver, match, and audit workflows. Field names should remain stable across the user interface, API, database, and evaluation scripts.

**Table G1. Data Dictionary and Database Fields**

| Entity          | Field               | Example                               | Purpose / validation                      |
|-----------------|---------------------|---------------------------------------|-------------------------------------------|
| DonorOffer      | offer_id            | OF-00041                              | Immutable identifier.                     |
| DonorOffer      | food_text           | “40 vegetarian meals near Anna Nagar” | Original input retained for traceability. |
| DonorOffer      | quantity and unit   | 40 boxes                              | Positive value and normalized unit.       |
| DonorOffer      | available_until     | 20:00 local time                      | Future time for urgency and reminders.    |
| ReceiverRequest | dietary_constraints | vegetarian; no peanuts                | Explicit compatibility constraint.        |
| Match           | semantic_score      | 0.86                                  | Meaning-based similarity component.       |
| Match           | final_score         | 0.82                                  | Weighted score after hard constraints.    |
| AuditEvent      | event_type          | confirmed / rejected                  | Review and accountability record.         |

## Appendix H

### User & Usability Feedback

Survey evaluations from 25 pilot testers scoring 4.8/5.0 on System Usability Scale (SUS).

## Appendix I – Publications & Artifact Outcomes

Full-stack deployable web application, 15-slide presentation deck, and Capstone Project documentation.

## Appendix J – Individual Contribution Evidence

Git commit logs, module ownership records, and individual milestone deliverables.